

AI Teaching Assistants in Asynchronous Discussion Forums: A Mixed-Methods Study of Students' Community of Inquiry Presence

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Abstract

Grounded in the Community of Inquiry (CoI) framework, this study examined how an artificial intelligence teaching assistant (AI TA) shaped students' perceived and behaviorally enacted CoI presence in asynchronous discussion forums (ADFs). TA support was examined in a 10-week blended course through a three-stage experimental cohort design, including no-TA, human-TA-supported, and AI-TA-supported discussion stages, together with a historical control cohort receiving continuous human-TA support for behavioral comparison. An explanatory mixed-methods design was adopted, combining CoI questionnaires, coded student-authored forum posts, forum-based reflections, and semi-structured interviews. The results revealed a perception-behavior gap: students reported higher cognitive presence (CP), social presence (SP), and teaching presence (TP) during the AI-TA-supported stage, but these perceived improvements were not fully mirrored in observable forum behaviors. Behaviorally enacted CP increased most clearly during the human-TA-supported stage in the experimental cohort and showed no further significant increase during the AI-TA-supported stage, while behaviorally enacted SP and

learner-enacted TP remained sparse in student-authored posts. Qualitative findings suggested that, compared with human TA support, the AI TA was perceived as advantageous in providing structured cognitive support, a low-pressure interaction atmosphere, and timely and accessible guidance, but limited in inquiry flexibility, emotional authenticity, and supervisory authority. These findings highlight the role boundaries of AI TAs in ADFs: they may strengthen students' perceived support, but their contribution to deeper inquiry and peer interaction depends on human orchestration and complementary facilitation. AI TAs should therefore be integrated into human–AI collaborative facilitation models that preserve the complementary roles of human TAs and instructors.

Keywords: artificial intelligence teaching assistant; asynchronous discussion forums; Community of Inquiry; pedagogical agents; mixed-methods

Introduction

Asynchronous discussion forums (ADFs) serve as a cornerstone of online and blended learning, with wide application in Massive Open Online Courses (MOOCs) and small private online courses. By fostering learning communities and providing interactive educational experiences, ADFs enable students to develop disciplinary expertise and critical thinking skills through engagement with peers and instructors (Robbins & Fairbank, 2023). The asynchronous nature of these forums provides learners with ample time for reflection (Hull & Saxon, 2009), allowing them to formulate inquiries, craft considered responses in online discussion (Miyashita & Wark, 2024), and engage in collaborative knowledge construction and critical thinking (Galikyan & Admiraal, 2019). However, due to the finite time and energy of instructors, providing timely and

meaningful replies to the vast volume of posts is a significant challenge. Coupled with the inherent latency in peer interactions, students often experience insufficient discussion feedback, which may negatively affect their learning experience and outcomes. While prior research has attempted to mitigate this through strategies such as setting up teaching assistants (Yang, 2008), optimizing instructor-student ratios (Parks-Stamm et al., 2017), and implementing reply-priority systems (Irish et al., 2023), these approaches rely heavily on limited human resources and cannot fully resolve the need for timely and sustained facilitation in ADFs.

Pedagogical Agents (PAs) offer a potential approach to addressing these challenges. They can be designed with course-specific knowledge and theory-informed interaction logic to provide timely feedback in online learning environments. A few studies have deployed agents as teaching assistants (TAs) on online learning platforms. For example, Khanmigo, an AI-powered tutoring chatbot developed by Khan Academy, offers Socratic guidance to students while assisting teachers with instructional tasks (Khan Academy, *n.d.*). Another example is Jill Watson, an AI virtual teaching assistant (Design Intelligence Lab, *n.d.*). It can answer inquiries regarding course syllabi and engage students in meaningful conversations about course materials (Taneja et al., 2024).

Although ADFs serve as a cornerstone of online and blended learning and support reflection, peer interaction, and collaborative knowledge construction, the application of PAs in ADFs remains underexplored. In particular, existing PAs have primarily been used for individual tutoring and platform-based learning support (Apoki et al., 2022; Martha & Santoso, 2019), whereas their role in facilitating ongoing peer discussion in ADFs has received limited attention.

This study addresses this gap by designing a PA that functions as a teaching assistant (TA) within ADFs and participates in the ongoing discussion process. Given that cognitive presence (CP), social presence (SP), and teaching presence (TP) are considered key indicators of meaningful online learning and substantially influence students' learning outcomes in online and blended learning environments (Martin et al., 2022), the Community of Inquiry (CoI) framework was adopted to guide the design and evaluation of the AI TA. The study then employed an explanatory mixed-methods design to examine students' self-reported CoI presence, behaviorally enacted CoI presence in forum interactions, and perceptions of the AI TA under different TA-supported discussion conditions.

Literature Review

CoI Presence in Online Discussion Forums

The CoI framework argues that cognitive, social, and teaching presences interactively support meaningful learning processes (Garrison et al., 1999; see Fig. 1). In online education, it provides theoretical guidance for the design of learning environments and the evaluation of learning experiences (Akyol & Garrison, 2008). Higher perceived levels of the three presences have been associated with better course grades, student experience, perceived learning, and satisfaction (Rockinson-Szapkiw et al., 2016; Stenbom, 2018).

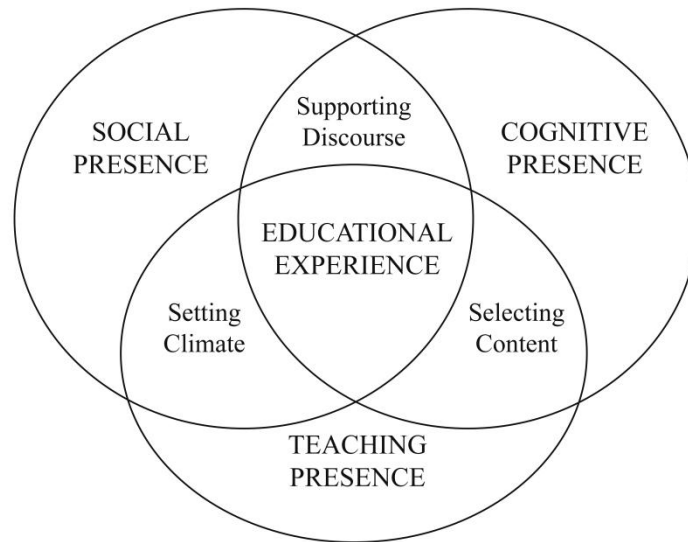


Fig. 1 Community of Inquiry framework. Adapted from Garrison et al. (1999).

.CP refers to the degree to which learners “collaboratively construct meaning through sustained discourse within a CoI” (Garrison et al., 2001). It is commonly analyzed through the Practical Inquiry Model, including triggering events, exploration, integration, and resolution (Garrison et al., 1999). A persistent challenge in asynchronous discussions is that cognitive activity often stalls at the exploration phase, rarely reaching higher-order integration or resolution (Garrison & Arbaugh, 2007; Naeem & Khan, 2019). This suggests that merely providing an online space for discussion is insufficient; students need strategic prompts, questioning, synthesis, and guidance to move from information exchange toward deeper inquiry (Darabi et al., 2011; Koh et al., 2010).

SP is defined as the “capacity of participants in a CoI to project themselves socially and emotionally as ‘real’ people” (Garrison et al., 2010). It serves as the foundation for a collaborative learning community by supporting affective expression, open communication, and group cohesion. In ADFs, SP is especially important because students interact without immediate

face-to-face cues and must rely on textual communication to establish interpersonal connection, trust, and group belonging. However, sustaining SP remains challenging when peer responses are delayed, participation is uneven, or students perceive the forum as a task-completion space rather than a learning community.

TP involves the “design, facilitation, and direction of cognitive and social processes” to achieve meaningful learning outcomes (Anderson et al., 2001). It is enacted through instructional design, discourse facilitation, direct instruction, and activities such as clarifying expectations, encouraging participation, maintaining discourse, asking probing questions, and providing feedback (Garrison, 2007). Although TP is often initiated by instructors, it may also be distributed among instructors, teaching assistants, and students during online discussions. However, because instructors and human teaching assistants have limited time and attention, providing timely and sustained facilitation for every student post remains difficult, especially in forums with large volumes of discussion.

In summary, the CoI framework underscores that ADFs can support reflection, peer interaction, and collaborative knowledge construction, but these outcomes depend heavily on effective facilitation. These challenges call for technology-supported facilitation in CoI-oriented ADFs.

Pedagogical Agents in Online Learning

Pedagogical Agents (PAs) are defined as life-like characters or software interfaces designed to facilitate learning in computer-mediated environments (Kim & Baylor, 2006). Traditionally, PAs served as demonstrators or tutors within structured learning systems, focusing on delivering

content or guiding students through predefined learning paths. With the integration of natural language processing and large language models (LLMs), PAs have gradually shifted from scripted and presentation-oriented agents to more conversational, adaptive, and generative learning partners (Lee et al., 2024). A recent systematic review shows that AI chatbots can provide personalized learning support, immediate feedback, homework assistance, skill development, and time-saving pedagogical assistance, while also raising concerns about accuracy, reliability, transparency, ethical use, and human oversight (Labadze et al., 2023).

In online learning environments, AI agents and PAs have been used to support several instructional and interactional functions. First, they can serve as tutors or content presenters by explaining concepts, modeling procedures, and guiding learners' attention to key information (Zhang et al., 2024). Second, AI agents can provide feedback, scaffolding, and formative guidance during learning activities (Labadze et al., 2023). Studies of educational chatbots and online discussion agents show their potential to provide immediate, personalized, and contextually coherent support that helps learners interpret information, monitor progress, and participate in online learning activities (Li et al., 2022).

Third, AI agents may support affective and social dimensions of online learning. Affective and chatbot-based agents can provide emotional support, immediate responses, and social cues (Labadze et al., 2023), while empathic chatbot research suggests that machine expressions of sympathy and empathy can shape users' perceptions of social and emotional support (Liu & Sundar, 2018). Finally, AI agents can be embedded in broader platform-based learning environments to support multiple roles, including teaching, peer interaction, co-regulation, and

learning guidance, but such support still requires careful human-centered design and oversight (Ouyang & Jiao, 2021; Zawacki-Richter et al., 2019).

Taken together, existing studies indicate that AI agents can support online learning through tutoring, feedback, scaffolding, affective support, learning guidance, and platform-based interaction. However, much of this work has focused on individual learner-agent interaction, video-based instruction, formative assessment, or general platform support. It is underexplored how an AI agent can support learning in asynchronous discussion forums, where learning depends not only on individual support but also on sustained peer interaction, discourse facilitation, and collaborative inquiry.

Toward CoI-Oriented AI TA Facilitation in ADFs

The functions of AI agents reviewed above suggest that they may be used to support the cognitive, social, and teaching presences that are central to CoI-oriented facilitation in ADFs. For CP, recent research demonstrates that GenAI can support students' cognitive engagement when used as a structured scaffold. For instance, Yu et al. (2025) integrated GenAI into six-hat thinking scaffolded discussions, where students consulted GenAI for information retrieval, idea generation, and problem-solving during instructional design tasks. Their findings suggested that GenAI-supported groups generated more discussion posts and showed stronger connections among higher-level CP indicators, particularly among high-creativity pre-service teachers. However, in that study, GenAI functioned mainly as a student-initiated cognitive tool rather than an embedded facilitator that directly participated in forum interactions.

In terms of SP, AI assistants may provide human-like social cues at scale. Research on AI

interfaces and customer-service chatbots suggests that anthropomorphic design cues and immediate responses can enhance user perceived social presence and improve their attitudes toward the interface or brand (Rao Hill & Troshani, 2024; Williams, 2025). However, these chatbots mainly function as one-to-one service or interface agents, aiming to create interpersonal warmth, responsiveness, and user engagement, rather than to facilitate peer interaction or group discourse in collaborative learning communities.

For TP, AI agents may also enact certain facilitation and instructional support functions. Wang et al. (2024) found that learners perceived cognitive, social, and teaching presences when interacting with an AI coach, and that teaching presence negatively predicted learning outcomes. They suggested that this result may reflect limitations in the design and implementation of AI support rather than the inherent inability of AI agents to facilitate learning. This finding is especially relevant for ADFs because AI-based teaching support may be perceived by students as useful, but such perceptions do not necessarily indicate that AI support effectively enhances sustained inquiry and learning outcomes.

Together, prior studies suggest that AI agents may provide social cues (Rao Hill & Troshani, 2024; Williams, 2025), cognitive scaffolding (Yu et al., 2025), and instructional guidance for online learning (Li et al., 2022; Wang et al., 2024), while CoI-oriented ADF research has mainly fostered online discussion through human facilitation strategies, such as instructor modeling and discussion prompts (Darabi et al., 2011; Koh et al., 2010), feedback and participation requirements (Yang, 2008), and teaching assistant support (Yang, 2008). Embedding an AI TA in ADFs differs from using AI chatbots for individual tutoring, customer-service interaction, or

student-initiated cognitive support. ADFs are collaborative learning spaces (Gao et al., 2013) in which knowledge construction depends on sustained peer interaction (Gunawardena et al., 1997), shared inquiry (Fu et al., 2016), and the progressive development of discussion threads (Hewitt, 2003). Although emerging studies on GenAI tools and pedagogical agents suggest the potential of AI-supported facilitation to provide social cues (Rao Hill & Troshani, 2024; Williams, 2025), cognitive scaffolding (Yu et al., 2025), and instructional guidance (Wang et al., 2024), few studies have examined whether such agent-based support can be embedded into ADFs to facilitate the three dimensions of CoI presence during ongoing discussion.

Research Gap and Present Study

Prior CoI research has established CP, SP, and TP as interrelated constructs for understanding meaningful online and blended learning experiences (Redstone et al., 2018), while research on PAs, especially pedagogical AI conversational agents, has shown their potential to provide cognitive guidance, affective support, feedback, and learning support in online learning environments (Yusuf et al., 2025). However, these two research streams remain insufficiently connected in ADFs. Limited evidence is available on how PAs, when embedded as AI TAs, participate in ADF interactions and shape students' CoI presence. To address this gap, the present study designed an AI TA for ADFs based on CoI-oriented facilitation principles and examined its role in students' CoI presence across different TA-supported discussion conditions. We evaluated AI TA support through students' self-reported and behavioral presences as well as their perceptions. The study was guided by the following research questions:

RQ1: How does students' self-reported CoI presence vary across no-TA,

human-TA-supported, and AI-TA-supported discussion stages?

RQ2: How does students' behaviorally enacted CoI presence vary across discussion stages and between the experimental cohort and the historical control cohort?

RQ3: What are students' overall perceptions of the AI TA, and how do they perceive its advantages and limitations compared with human TA support in ADFs?

Design and Implementation of the AI TA

This section describes the design and implementation of the AI TA as an embedded facilitator in ADFs, including its CoI-oriented interaction strategies and response-generation pipeline.

CoI-Oriented Interaction Design

Prior research suggests that instructional strategies effective for human instructors may also remain effective when implemented through PAs (Zhang et al., 2024). Therefore, the AI TA's interaction design was adapted from established facilitation strategies (Table 1). The AI TA was designed as a form of TA support comparable to human TA support in the study design.

These strategies were integrated into a response-composition logic rather than implemented as isolated response functions. For initial responses to student posts, the AI TA followed a four-step pattern to provide prompt feedback, acknowledge students' contributions, offer cognitive scaffolding, and invite further peer discussion. This pattern guided the composition of AI replies rather than prescribing how the discussion itself should unfold.

Specifically, the AI TA first established conversational immediacy through prompt feedback, peer identity, and friendly salutation. Second, it affirmed and synthesized students' contributions through content summarization, emotional support, and personalized guidance. Third, it provided

cognitive scaffolding through illustrative examples and critical challenges. Finally, the AI TA invited the original poster and other peers to respond to, extend, or build on one another's ideas.

The four-step pattern was mainly applied to the AI TA's initial responses to student posts. In subsequent multi-turn conversations, the AI TA generated more concise and context-sensitive replies, reducing scaffolding while retaining necessary emotional support. This design was intended to avoid overly lengthy or mechanical responses that might reduce students' agency, interfere with peer-to-peer interaction, or weaken continued engagement.

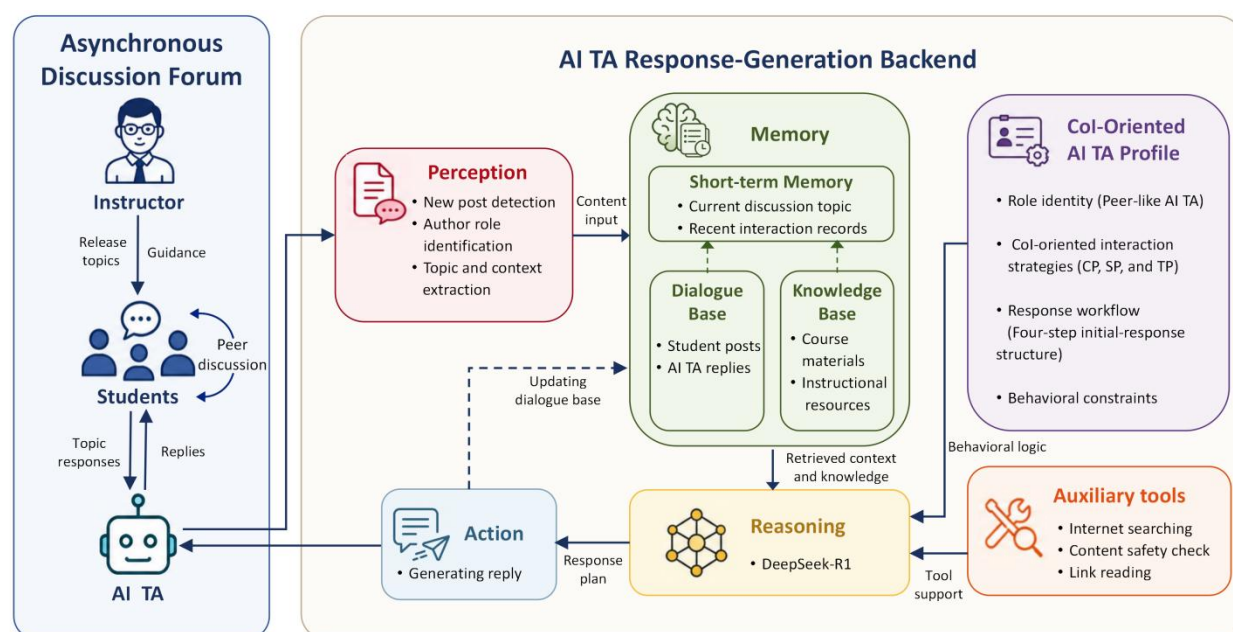
Table 1 CoI-oriented interaction strategies of the AI TA

Presence	Strategies	Description	Source
CP	Illustrative Scaffolding	Uses concrete examples to clarify concepts and help students elaborate their ideas	Garrison & Arbaugh (2007)
	Critical Challenge	Poses probing questions to stimulate reflection, perspective expansion, and higher-order inquiry	Darabi et al. (2011)
	Content Summarization	Summarizes key points from students' posts to support knowledge integration and discussion coherence	Garrison & Cleveland-Innes (2005)
	Peer Identity	Adopts a learning-companion identity that simulates human-peer-like interaction and supports socially oriented learning interaction	Kim & Baylor (2006)
SP	Friendly Salutation	Uses friendly greetings and emoticons to create affective warmth	Tang & Hew (2020)
	Emotional Support	Affirms students' contributions and provides encouragement to support continued participation	Scherer Bassani (2011)

	Prompt Feedback	Provides timely responses after students submit posts	Baker (2010)
TP	Personalized Guidance	Gives individualized suggestions based on students' posts and discussion context	Shea et al. (2003)
	Discussion Invitation	Invites students to respond to peers, extend discussion, and sustain interaction	Anderson et al. (2001)

System Implementation and Response Generation Process

The AI TA was developed on Coze¹, a generative AI agent development platform, and was connected to the ADFs through API-based integration. This integration allowed new student posts to be captured as input for backend processing and enabled generated replies to be returned through the AI TA forum account. As shown in Fig. 2, the system consisted of two connected components: the forum-side interaction workflow and the AI TA response-generation backend.



¹ Coze: <https://www.coze.com/>

Fig. 2 AI TA architecture and forum interaction workflow

On the forum side, the instructor initiated discussions by posting topics and guidance in the ADFs, while students participated by publishing original posts and replying to peers. When a new student post was submitted, it became visible to the AI TA and triggered the backend response-generation process. The generated response was posted back to the forum as the AI TA's reply. Instructor posts were treated as instructional prompts or guidance and did not trigger AI TA replies. In this way, the AI TA functioned as a visible forum-based teaching assistant, while the discussion space remained organized around instructor-initiated topics, student posts, and peer discussion.

On the backend side, response generation was organized through several functional modules. The perception module detected new posts, identified the author role, and extracted the post content, discussion topic, and immediate context. The memory module consisted of short-term and long-term memory. The short-term memory maintained conversational contexts, while the long-term memory included discussion log database and knowledge base. The discussion log database contained student posts and AI-generated replies, to support log retrieval and context updating. The knowledge base provided course materials and instructional resources. The profiling module specified the AI TA's identity setting, response workflow, and behavioral constraints. The reasoning module, powered by DeepSeek-R1, integrated the post input, recent context, retrieved discussion records, knowledge base, and the AI TA profile to generate a response plan. Auxiliary tools, including internet search, content safety checking, and link reading, were available when needed. Finally, the action module composed the final reply and

returned it to the forum through the AI TA account.

Interaction Example

An example of AI TA interaction in the ADFs is presented in Fig. 3.

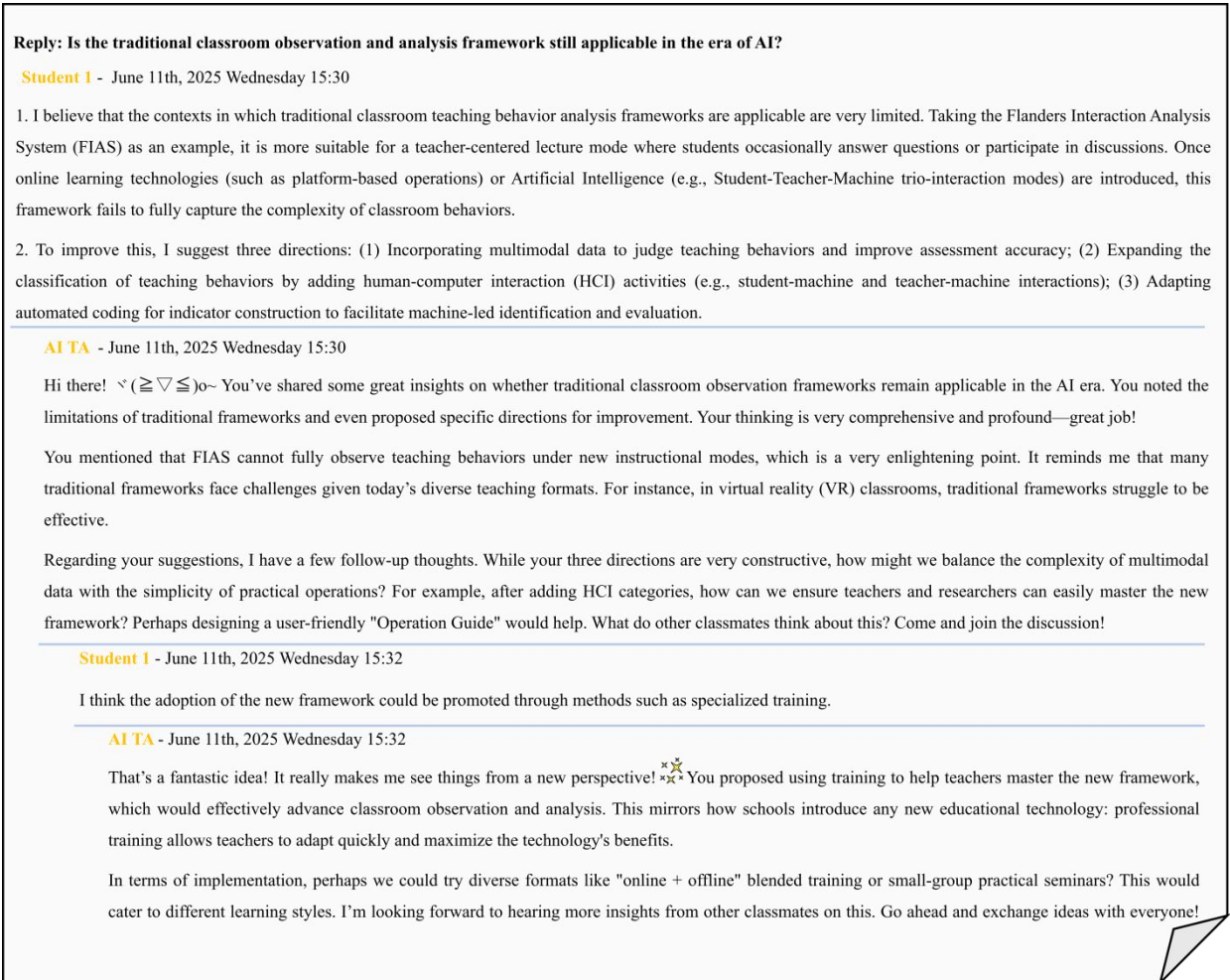


Fig. 3 Example of the AI TA interaction in ADFs

Methodology

This study employed an explanatory mixed-methods design (Fetters et al., 2013) with a quasi-experimental structure and a historical control cohort (Walser, 2014). Quantitative data were used to examine students' perceived and behaviorally enacted CoI presence across different

TA-support conditions, while qualitative data were used to further explain students' experiences with the AI TA and contextualize the quantitative findings.

Study Context and Participants

Data were collected from a blended graduate course, named Educational Technology, at a university in southern China. The course combined face-to-face classroom teaching with after-class online learning tasks, and ADF activities were integrated into the after-class learning.

The experimental cohort (2025 Spring) consisted of 59 first-year master's students majoring in educational technology, including 16 males and 43 females. All provided informed consent for their learning data to be used for research purposes. The historical control cohort (2023 Fall) comprised 50 first-year students enrolled in the same course, including 13 males and 37 females. Both cohorts were from the same program, shared comparable academic backgrounds and admission criteria, followed the same course design, and were taught by the same instructor. The historical cohort served as a comparison group under a human TA-supported condition.

Research Design and Data Collection

As shown in Fig. 4, the study followed a 10-week design combining a three-stage within-subject design in the experimental cohort with a historical control comparison. In the experimental cohort, students first participated in forum discussions without TA support during weeks 1–2, followed by human-TA-supported discussions during weeks 3–6 and AI-TA-supported discussions during weeks 7–10. Self-reported CoI surveys were administered at the end of each stage, and forum discussion posts were collected and coded to examine students' behaviorally enacted CoI presence. After the AI-TA-supported stage, students were asked to write reflection

posts about interaction experiences with the AI TA, and semi-structured interviews were conducted with six participants who were randomly selected from a convenience pool of students who were available and willing to participate.

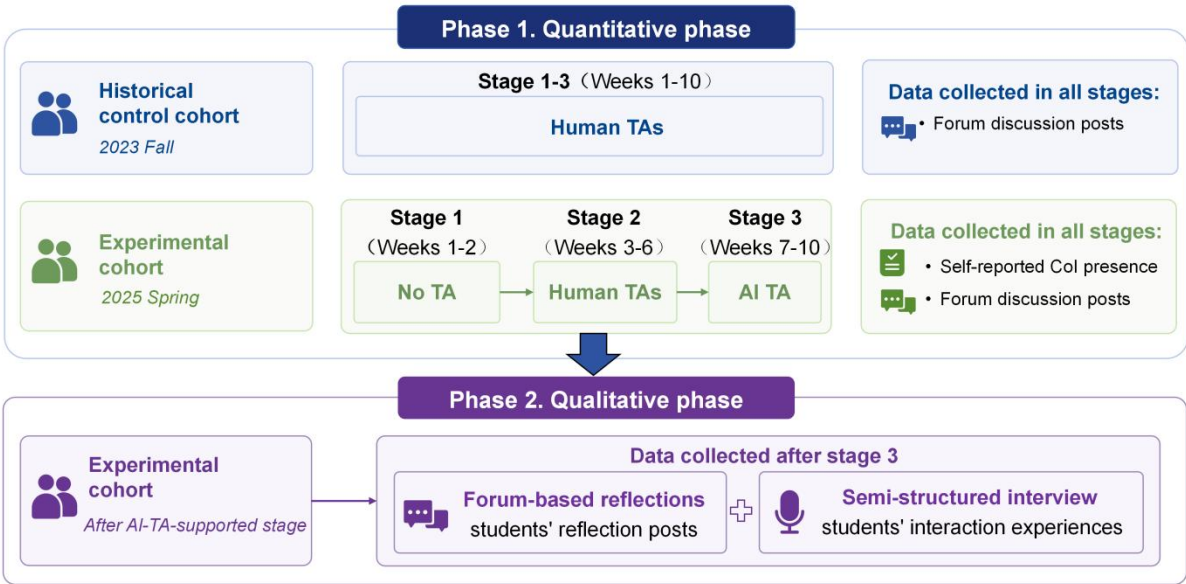


Fig. 4 Research design

The historical control cohort followed the same 10-week course structure and was supported exclusively by human TAs throughout the semester. Because the historical data were retrospective, only archived forum discussion posts were available. These posts were used as comparison data for examining students' behaviorally enacted CoI presence across TA-support conditions.

Measurements

Self-Reported CoI Presence

Students' self-reported CoI presence was measured using the Community of Inquiry questionnaire developed by Arbaugh et al. (2008), which was adapted into Chinese for the

present study. The instrument consists of 34 items rated on a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree, including 12 items for CP, 9 for SP, and 13 items for TP. Cronbach's α coefficients were calculated separately for each dimension at each discussion stage. As shown in Table 2, the coefficients ranged from 0.82 to 0.92, indicating good internal consistency.

Table 2 Cronbach's α coefficients for each CoI subscale across the three support stages

Dimension	No TA Stage	Human TA Stage	AI TA Stage
CP	0.86	0.89	0.91
SP	0.82	0.91	0.90
TP	0.85	0.90	0.92

Behaviorally Enacted CoI Presence

Students' behaviorally enacted CoI presence was examined through forum discussion posts collected across the discussion stages. The posts were analyzed using the CoI coding framework proposed by Garrison et al. (1999), which conceptualizes CP, SP, and TP through multiple discourse indicators. The operational definitions are shown in Table 3.

Table 3 Operational definitions of CoI presence indicators

Elements	Categories	Indicators (examples only)
CP	Triggering Event	Sense of Puzzlement
	Exploration	Information Exchange
	Integration	Connecting Ideas
	Resolution	Applying New Ideas
SP	Open Communication	Learning Climate/Risk-Free Expression
	Group Cohesion	Group Identity/Collaboration
	Personal/Affective	Self Projection/Expressing Emotions
TP	Design & Organization	Setting Curriculum & Methods Shaping
	Facilitating Discourse	Constructive Exchange
	Direct Instruction	Focusing and Resolving Issues

The behavioral CoI coding focused only on student-authored forum posts. Posts written by the instructor, human TAs, researchers, or the AI TA were treated as part of the instructional support conditions and were not included as coding units. This decision kept the behavioral analysis centered on students' enacted CoI presence and avoided conflating students' behavioral presence with the varying amount and form of facilitation provided under different TA-support conditions. To ensure comparability across stages and cohorts, the behavioral coding was based on selected forum tasks rather than all forum activities. Specifically, opinion discussion and learning task forums were included because they required students to articulate viewpoints, discuss course-related issues, or respond to instructional prompts. Forums serving mainly administrative or retrospective purposes, such as self-introduction forums and reflective summary forums, were excluded because they differed from regular discussion tasks.

To establish the reliability of the behavioral coding, two trained coders independently coded the forum posts according to the behavioral CoI coding scheme. Inter-coder agreement exceeded 80% across the coded categories, and discrepancies were resolved through discussion until consensus was reached before the final coding results were used for analysis.

For CP, student posts were coded according to the four phases of the Practical Inquiry Model: triggering event, exploration, integration, and resolution (Garrison et al., 2001). In addition to phase frequency, a weighted behavioral CP score was calculated to distinguish lower-level inquiry from higher-order meaning construction, as online discussions often remained at the exploration phase and less frequently progressed to integration or resolution (Garrison & Arbaugh, 2007; Garrison & Cleveland-Innes, 2005; Naeem & Khan, 2019).

Triggering event and exploration were assigned a weight of 1, whereas integration and resolution were assigned a weight of 2. This scoring strategy retained the frequency-based logic of traditional analyses based on the Practical Inquiry Model while introducing a distinction between lower-level and higher-level inquiry:

$$CP = \frac{1 \times n_{triggering} + 1 \times n_{exploration} + 2 \times n_{integration} + 2 \times n_{resolution}}{n_{forums}}$$

For SP, because the selected forums were mainly topic-based opinion discussions and learning tasks, a strict coding standard was adopted to avoid overestimating SP from task-induced personal statements. Expressions of personal experience, preference, emotion, or attitude were coded as SP only when they explicitly functioned as interpersonal interaction, such as acknowledging another participant, responding to a peer's idea, expressing appreciation, and building a sense of group connection. In contrast, first-person statements that merely described one's own learning experience, viewpoint, preference, and task-related reflection were not coded as SP. This distinction was intended to ensure that behavioral SP reflected interpersonal and social interaction rather than the mere presence of personal pronouns or affective wording.

For TP, student posts were coded only when they reflected relevant behaviors, such as organizing discussion, facilitating peer exchange, focusing issues, or guiding collective inquiry. TP provided by instructors, human TAs, or the AI TA was excluded.

Qualitative Data on Students' Perceptions of the AI TA

To address RQ3, qualitative data were collected after the AI TA-supported stage through forum-based reflections and semi-structured interviews. In the reflection activity, the instructor

initiated a thread with the prompt: “How do you perceive the performance of the AI TA? Has she helped you? What needs to be improved?” This activity yielded 32 reflection posts from the experimental cohort.

Semi-structured interviews were conducted with six participants, including four males and two females, to obtain more in-depth insights into students’ interaction experiences with the AI TA and their perceived advantages and limitations of AI TA support relative to human TA support across the three CoI dimensions. The interview protocol was guided by the CoI framework and focused on students’ perceptions of AI TA support for guidance and facilitation, social connection and participation, as well as thinking and understanding. It also included questions asking participants to compare AI TA support with human TA support in terms of timeliness, accessibility, interaction pressure, emotional support, supervisory authority, and inquiry flexibility. The reflection posts and interview transcripts were then used together for qualitative analysis.

Data Analysis

Self-Reported CoI Presence

For the self-reported CoI questionnaire data, only students who completed all three stage-specific surveys were included ($n = 40$). After Bonferroni correction for nine Shapiro–Wilk tests, only CP and SP scores in the no-TA stage deviated from normality (adjusted $p = .008$ and $.009$); the remaining distributions did not (adjusted $ps \geq .516$). Given the robustness of repeated-measures ANOVA to moderate non-normality, the analyses were retained (Blanca et al., 2017). Sphericity was met for CP and TP but violated for SP; therefore, Greenhouse–Geisser correction was

applied to SP ($\epsilon = .680$; Blanca et al., 2024). Bonferroni-adjusted pairwise comparisons were conducted.

Behaviorally Enacted CoI Presence

Based on the forum selection criteria described in Section 4.3.2, six comparable forum tasks were retained for the experimental cohort, with two from each stage, and three tasks for the historical control cohort, with one from each stage. After excluding instructor prompts, human-TA/researcher feedback, AI-TA replies, and other non-student posts, the behavioral corpus included 300 student-authored posts from the experimental cohort and 154 from the historical control cohort. The full roster-based samples were retained, including 59 experimental and 50 historical-control students. If a student did not contribute analyzable posts in a selected forum or stage, the corresponding behavioral score was coded as zero.

After coding and aggregation, CP showed sufficient variation across students and stages, whereas SP and TP showed highly sparse distributions. Therefore, inferential analyses were conducted only for behavioral CP, while SP and TP were reported using descriptive statistics. Shapiro–Wilk tests indicated that behavioral CP scores significantly deviated from normality across stages and cohorts ($W = .71-.92$, $ps \leq .001$). Because the scores were discrete, included zero values, and did not meet the normality assumption, a rank-based nonparametric longitudinal analysis was conducted using the nparLD package in R (Noguchi et al., 2012). An F1-LD-F1 model was specified, with Stage (S1, S2, and S3) as the within-subject factor and Cohort (experimental vs. historical control) as the between-subject factor. Rank-based ANOVA-type statistics (ATSSs) were used to examine the main effects of Stage and Cohort and the Stage $\times$

Cohort interaction. Following a significant interaction, within-cohort stage differences were examined using Friedman tests, followed by paired Wilcoxon signed-rank tests with Bonferroni-adjusted p values. Stage-specific differences between cohorts were examined using Wilcoxon rank-sum tests with Bonferroni adjustment. Relative treatment effects (RTEs) were reported as rank-based effect estimates.

Qualitative Analysis of Students' Perceptions of the AI TA

Students' perceptions of the AI TA were analyzed using two qualitative procedures. First, the 32 forum-based reflection posts were first examined through attitude classification and keyword frequency analysis. Responses were manually categorized as positive, neutral, or negative according to students' affective stance toward the AI TA. Frequently mentioned features were identified using a dictionary-based Bag-of-Words document-frequency analysis implemented in Python with scikit-learn (Juluru et al., 2021; Pedregosa et al., 2011). Semantically related words and phrases were grouped into keyword categories, and each category was counted once per post.

Then, the reflection posts and semi-structured interview transcripts were analyzed thematically following Braun and Clarke's (2006) six-step framework: familiarization, initial coding, theme searching, theme reviewing, theme defining, and reporting. The thematic analysis focused on students' overall perceptions of the AI TA and their perceived advantages and limitations of AI TA support relative to human TA support. The analysis was primarily conducted by one researcher. To enhance trustworthiness, the researcher repeatedly read the reflection posts and interview transcripts, generated initial codes, compared patterns across the

two qualitative data sources, and iteratively refined candidate themes by checking them against the original data. The coding structure, candidate themes, and representative quotations were then reviewed and discussed with the research team to improve analytic coherence. Representative quotations were translated from Chinese into English by the authors and checked against the original transcripts to preserve their intended meaning.

Results

Quantitative Findings

Perceived CoI presence at different stages

Table 4 presents the descriptive statistics and repeated-measures ANOVA results for students' self-reported CP, SP, and TP across the three support stages. Significant stage effects were found for CP ($F_{2,78} = 79.40, p < .001$, partial $\eta^2 = .67$), SP ($F_{1.36, 53.06} = 50.79, p < .001$, partial $\eta^2 = .57$), and TP ($F_{2,78} = 150.79, p < .001$, partial $\eta^2 = .80$).

Table 4 Descriptive statistics and repeated-measures ANOVA results for perceived CoI presence

Dimension	Mean (SD)			F	p	Partial η^2	Post-hoc comparison
	No TA	Human TA	AI TA				
CP	3.59 (0.46)	3.74 (0.42)	3.97 (0.54)	79.40	< .001	.67	AI > Human > No
SP	3.53 (0.63)	3.64 (0.45)	3.94 (0.54)	50.79†	< .001	.57	AI > Human = No
TP	3.66 (0.54)	3.69 (0.46)	4.04 (0.52)	150.79	< .001	.80	AI > Human = No

Note. † indicates that the Greenhouse-Geisser correction was applied. Pairwise comparisons were adjusted using the Bonferroni correction.

For CP, all pairwise comparisons were significant, with the highest score in the AI-TA-supported stage, followed by the human-TA-supported and no-TA stages. For SP and TP, students reported significantly higher scores in the AI-TA-supported stage than in both the

human-TA-supported and no-TA stages, while the human-TA-supported and no-TA stages did not differ significantly.

Behaviorally Enacted CoI Presence Across Cohorts and Stages

Behavioral analyses of coded forum discussion posts revealed more limited changes in students' interaction behaviors. CP showed sufficient variation for inferential analysis, whereas behaviorally enacted SP and TP appeared only sparsely in student-authored posts. Therefore, inferential analyses were conducted only for behavioral CP, while SP and TP were analyzed with descriptive statistics.

Cognitive Presence. As shown in Fig. 5, students' behaviorally enacted CP showed different developmental patterns across the experimental and historical control cohorts. In the experimental cohort, the average behavioral CP score per comparable forum task increased from S1 ($M = 1.03$, $SD = 0.33$) to S2 ($M = 1.35$, $SD = 0.60$), followed by a slight decrease in S3 ($M = 1.14$, $SD = 0.71$). The corresponding RTEs were .418, .583, and .488, respectively. In the historical control cohort, CP scores showed no clear increase across stages, with S1 ($M = 1.44$, $SD = 1.03$), S2 ($M = 1.20$, $SD = 0.57$), and S3 ($M = 1.16$, $SD = 0.84$). The corresponding RTEs were .544, .498, and .471, respectively.

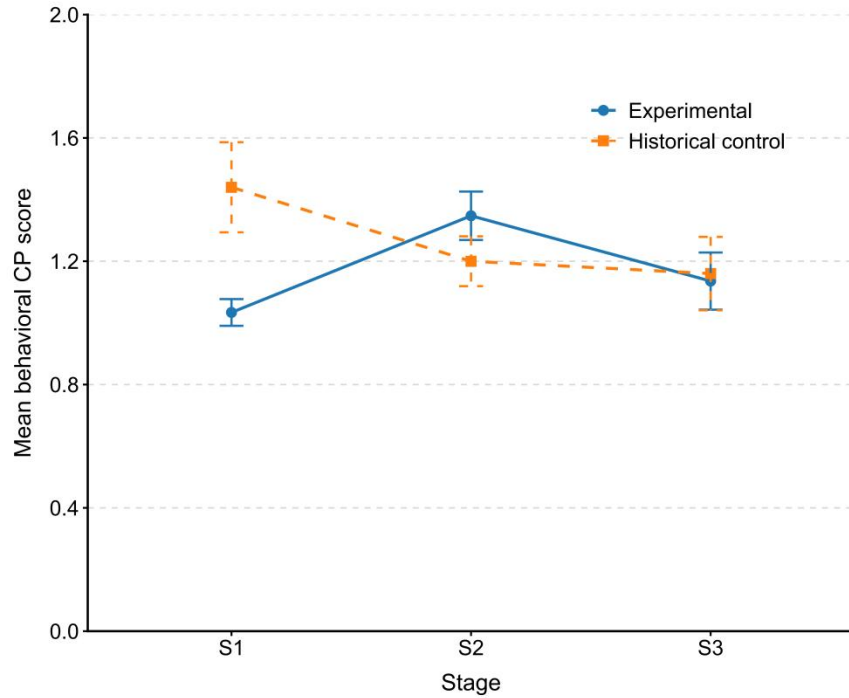


Fig. 5 Mean behavioral CP scores by cohort and stage

At the post level, the experimental cohort showed a shift from predominantly exploratory discourse in S1 to more resolution-oriented discourse in S2, followed by a partial return to exploratory discourse in S3. Specifically, exploration accounted for 83.49% of student posts in S1, decreased to 39.00% in S2, and increased again to 51.65% in S3, whereas resolution increased from 11.93% in S1 to 57.00% in S2 and then decreased to 43.96% in S3. In the historical control cohort, exploration remained the most common CP category across stages. The full distribution of student posts across CP categories is shown in Table 5.

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Table 5 Distribution of student posts across CP categories by cohort and stage

Cohort	Stage	Triggering Event	Exploration	Integration	Resolution	None
Experimental	S1	3 (2.75%)	91 (83.49%)	1 (0.92%)	13 (11.93%)	1 (0.92%)
	S2	0 (0.00%)	39 (39.00%)	3 (3.00%)	57 (57.00%)	1 (1.00%)
	S3	1 (1.10%)	47 (51.65%)	3 (3.30%)	40 (43.96%)	0 (0.00%)
Historical Control	S1	0 (0.00%)	34 (62.96%)	1 (1.85%)	18 (33.33%)	1 (1.85%)
	S2	0 (0.00%)	40 (78.43%)	5 (9.80%)	5 (9.80%)	1 (1.96%)
	S3	0 (0.00%)	26 (53.06%)	4 (8.16%)	12 (24.49%)	7 (14.29%)

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The rank-based nonparametric longitudinal analysis showed no significant main effect of Stage, $ATS = 2.06$, $df = 1.94$, $p = .129$, and no significant main effect of Cohort, $ATS = 0.05$, $df = 1.00$, $p = .815$. However, the Stage $\times$ Cohort interaction was significant, $ATS = 4.99$, $df = 1.94$, $p = .007$, indicating that the developmental pattern of behaviorally enacted CP differed between the two cohorts.

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Follow-up Friedman tests showed that behavioral CP differed significantly across stages in the experimental cohort, $\chi^2(2) = 11.49$, $p = .003$, but not in the historical control cohort, $\chi^2(2) = 2.63$, $p = .269$. Bonferroni-adjusted paired Wilcoxon signed-rank tests indicated that, in the experimental cohort, CP was significantly higher in S2 than in S1 ($V = 251.00$, adjusted $p = .002$), whereas S1 and S3 (adjusted $p = .858$) and S2 and S3 (adjusted $p = .208$) did not differ significantly. None of the within-cohort stage comparisons was significant in the historical control cohort (adjusted $ps \geq .383$). Bonferroni-adjusted Wilcoxon rank-sum tests further showed that the historical control cohort had significantly higher CP than the experimental cohort in S1 ($W = 1123.50$, adjusted $p = .044$), whereas the cohorts did not differ significantly in S2 or S3

(adjusted p s $\geq .379$). Taken together, the significant interaction primarily reflected an increase from S1 to the human-TA-supported S2 in the experimental cohort, followed by no further significant change in the AI-TA-supported S3, while the historical control cohort remained relatively stable across stages.

Social Presence. Behaviorally enacted SP was rarely observed. In the experimental cohort, no student-authored posts were coded as containing SP indicators across the three stages. In the historical control cohort, only a few posts contained explicit SP indicators: 1 of 54 posts in S1 (1.85%), 2 of 51 posts in S2 (3.92%), and 2 of 49 posts in S3 (4.08%). These indicators mainly involved explicit peer acknowledgement, appreciation, or alignment with another participant's idea or experience. Overall, students' observable SP behaviors were limited.

Teaching Presence. Behaviorally enacted TP was not observed in student-authored posts. No student posts were coded as design and organization, facilitating discourse, or direct instruction. This suggests that TP in the selected forums was mainly enacted by instructors, human TAs, or the AI TA. Thus, although students reported higher perceived TP during the AI-TA-supported stage, this perception was not reflected in their behaviors.

Qualitative Findings

The qualitative findings provide further insight into students' perceptions of the AI TA and their interaction experiences in ADFs.

Students' Overall Perceptions of the AI TA

The forum-based reflection posts showed a predominantly positive reception of the AI TA. Among the 32 reflection posts, 28 students (87.50%) expressed positive attitudes, mainly

praising the AI TA's guiding questions, encouraging tone, and timely responses. Three students (9.40%) expressed neutral views, describing the AI TA as a functional tool with limited influence on their discussion habits. One student (3.10%) expressed a critical view, focusing on the relatively low information quality of the AI TA's feedback.

Table 6 High-frequency keywords from forum-based reflections

Rank	Keywords	Frequency
1	Timely / Fast	22
2	Guiding / Inspiring	22
3	Encouraging / Affirming	21
4	Criticality / Questioning	17
5	Patterned	13
6	Friendly / Emojis	12
7	Summarizing / Sorting	11
8	Deep thinking	7
9	Emotional value	4
10	Liberating / reduction	3

To further identify salient features in students' reflections, a keyword frequency analysis was conducted. As shown in Table 6, the most frequently mentioned terms were related to guidance, critical questioning, summarization, encouragement, and timely responses. Students also mentioned patterned or repetitive interaction styles, suggesting concerns about limited conversational flexibility. Taken together, the reflection posts suggest that students primarily perceived the AI TA as a responsive and supportive facilitator, whose support was sometimes formulaic or insufficiently adaptive.

Perceived Advantages and Limitations of the AI TA Compared with Human TA

The thematic analysis based on forum-based reflection posts and semi-structured interviews identified two overarching categories (Table 7): students' perceived advantages and limitations of the AI TA compared with human TA support.

Table 7 Thematic mapping results

Categories	Themes	Representative initial codes
Perceived advantages	Structured cognitive support	Rapid summarization; organizing ideas; perspective expansion; critical questioning; guiding new ideas; reduced cognitive load
	Low-pressure interaction atmosphere	Anthropomorphic approachability; reduced psychological barriers; friendly greetings; emotional encouragement
	Timely and accessible guidance	Overcoming temporal constraints; 24/7 guidance; fast responses; personalized responses
Perceived limitations	Limited inquiry flexibility	Predictable response structures; repetitive interaction logic; fixed replying patterns
	Limited emotional authenticity	Patterned praise; mechanical emotional responses; deficit in emotion regulation
	Lack of social authority	Lack of authoritative pressure; limited sense of accountability; reluctance to post follow-up responses

Perceived Advantages. Students generally perceived the AI TA as advantageous compared with human TA support in three ways: structured cognitive support, a low-pressure interaction atmosphere, and timely and accessible guidance.

Theme 1: Structured cognitive support. Students valued the AI TA's cognitive support, including summarization, idea organization, perspective expansion, and critical questioning. One

participant stated that the AI “helps me sort out the logic of my replies and summarize my answers”. Others noted that the AI TA could provide new questions and directions after evaluating their replies, which encouraged reflection and perspective expansion. These comments suggest that the AI TA helped students organize ideas and extend their thinking during asynchronous discussion.

Theme 2: Low-pressure interaction atmosphere. Students felt that the AI TA created a lower-pressure interaction atmosphere than did the human TA. They frequently mentioned the AI TA’s “friendly greetings” and “cute emoticons”, and several participants described it as more approachable than a human TA. One participant stated that the AI TA “fully affirms our replies every time, delivering top-tier emotional support”. These responses made interaction feel less stressful and more encouraging, partly because the AI TA did not carry the same evaluative pressure as a human facilitator.

Theme 3: Timely and accessible guidance. Compared with human TA support, students especially appreciated its immediacy, availability, and consistency. One participant noted that the AI TA “has a very fast response speed and can respond quickly at any time”, while another mentioned that it could “carry out personalized and targeted replies based on my comments”. These comments suggest that the AI TA mitigated the issue of delayed feedback in traditional ADFs.

Perceived Limitations. Despite these advantages, students also identified three main limitations of the AI TA: limited inquiry flexibility, limited emotional authenticity, and lack of supervisory authority.

Theme 4: Limited inquiry flexibility. Students perceived the AI TA's inquiry support as insufficiently flexible. One participant remarked that "the way she replies is relatively fixed, and the reply content and ideas lack variety". Others described the responses as increasingly routine over time. These comments suggest that while the AI TA provided useful prompts and structured support, its repetitive response logic limited students' sense of dynamic and adaptive interaction.

Theme 5: Limited emotional authenticity. Students questioned the AI TA's emotional authenticity. Some described its praise as "stereotyped" or mechanical, suggesting that repeated affirmation could feel superficial rather than genuinely understanding. One student explained that the AI had advantages in "emotional guidance, but not in emotional regulation". Thus, while the AI TA offered emotional support, students perceived it as less useful than human TA's emotional support.

Theme 6: Limited perceived authority. Students perceived the AI TA as having limited social authority within the course. One participant stated that the AI "has no way to give you the sense of accountability like a real person... you just glance at its reply and pass it". Several students similarly reported low motivation to post follow-up responses after receiving AI feedback as they felt little obligation to continue the interaction. Unlike human instructors or TAs, whose comments implicitly carry expectations for acknowledgment or response, the AI TA was not perceived as an authority figure to whom students were accountable. As a result, while the AI TA lowered the psychological pressure associated with participation, its limited perceived authority also weakened students' sense of accountability to sustain meaningful interactions.

Discussion

This study employed a mixed-methods design to examine how an AI TA shaped students' perceived and behaviorally enacted CoI presence in ADFs. The study linked students' self-reported CoI presence, observable forum behaviors, and perceptions of the AI TA's advantages and limitations compared with human TA support. For RQ1, students reported significantly higher self-reported CP, SP, and TP during the AI-TA-supported stage than during the no-TA or human-TA-supported stages. For RQ2, behavioral results showed a more limited pattern. Behaviorally enacted CP in the AI-TA-supported stage was higher than that in the no-TA stage but not different from the human-TA stage. SP was sparse in all stages, and learner-enacted TP was almost absent across stages. These findings suggest that although the AI TA was perceived as supportive, the present behavioral data provided limited evidence that this perceived support was accompanied by higher-order CP behaviors, stronger peer interaction, or learner-enacted facilitation. The findings revealed a perception-behavior gap, where the AI TA shaped students' CoI experience primarily by enhancing perceived support with limited effect on discussion behaviors under the current forum tasks.

For RQ3, compared with human TA, students generally perceived the AI TA positively, particularly in terms of structured cognitive support, critical questioning, low-pressure interaction, timeliness, and accessibility. However, they also identified AI TA's limitations in inquiry flexibility, emotional authenticity, and supervisory authority.

The Perception–Behavior Gap

For CP, the gap may stem from the AI TA’s structured but insufficiently adaptive cognitive support. Its summarization, idea organization, perspective expansion, and critical questioning likely made the discussion task feel clearer and more manageable. Such structured support can reduce unnecessary cognitive burden and strengthen learners’ sense that their thinking is being guided (Kasneci et al., 2023; Mollick & Mollick, 2023; Sweller, 1988), which helps explain the increase in perceived CP. However, higher-order CP in forum discourse requires more than receiving explanations or prompts. It demands sustained facilitation that elicits clarification, comparison of ideas, evidence-based reasoning, and application or resolution of ideas (Garrison & Cleveland-Innes, 2005). Students’ descriptions of the AI TA’s responses as relatively fixed suggest that its guidance may have functioned primarily as structured cognitive scaffolding rather than as adaptive discourse facilitation. This may explain why the AI TA enhanced perceived CP but showed limited effect in triggering integration- or resolution-level discourse.

For SP, the gap may stem from the AI TA’s low-pressure but emotionally limited social support. Its friendly tone, instant responsiveness, emoticons, and encouragement may have made discussion feel more approachable and less evaluative, thereby strengthening students’ perceived SP through greater immediacy and ease of interaction. Perceived SP is positively related to satisfaction and perceived learning (Richardson et al., 2017) and to students’ projection of SP indicators in online discussions (Swan & Shih, 2005). Recent studies on educational chatbots and conversational agents have found that such systems can provide immediate and personalized support (Labadze et al., 2023) and, in online discussion forums, can generate human-like,

contextually coherent, socio-emotionally supportive responses to support participation (Li et al., 2022). Emotional responsiveness and satisfaction with chatbot guidance have also been associated with favorable user perceptions and engagement-related outcomes (Liu & Sundar, 2018; Yildiz Durak, 2023). However, these perceptual gains did not necessarily indicate stronger peer-to-peer interaction. Observable SP requires students to enact social connection through affective expression, acknowledgement, reciprocal response, and group-oriented interaction (Rourke et al., 1999). Students' comments that the AI TA's praise was formulaic and emotionally inauthentic suggest that AI-generated encouragement may create a sense of friendliness without necessarily eliciting corresponding emotional expression or reciprocal social interaction among peers. Thus, the AI TA may have increased perceived SP by making interaction feel easier and more supportive, but this perceived social comfort was not accompanied with more frequent observable SP behaviors in forum discourse.

For TP, the gap may stem from the AI TA's visible but system-centered facilitation. Its 24/7 availability, immediate responses, and personalized guidance made instructional support more accessible, stable, and responsive, which likely strengthened students' perceived TP. This interpretation is consistent with CoI research showing that TP is closely related to learners' satisfaction and perceived online learning experience (Caskurlu et al., 2020). However, unlike perceived TP, learner-enacted TP requires students to actively assume facilitative responsibilities, such as organizing discussions, guiding peers, and sustaining collaborative inquiry. Enacting these roles also requires learners to exercise pedagogical judgment, assume responsibility for sustaining discussions, and adapt their facilitation to evolving interactions (Luckin & Cukurova,

2019; Molenaar, 2022). The AI TA summarized discussions, answered questions, and maintained interaction flow, which may have made facilitation feel well supported while reducing students' perceived need to assume these responsibilities themselves. Taken together, these findings suggest that AI may strengthen students' experience of TP without necessarily fostering learner-enacted TP. By taking over facilitative functions while lacking the social authority that elicits reciprocal engagement, the AI TA reduced students' need to organize, guide, and sustain discussions themselves.

Implications

This study contributes to AI-supported online learning research by showing how an AI TA shaped students' CoI presence in ADFs. Theoretically, it extends the CoI framework to AI-mediated discussion environments by showing that the AI TA enhanced perceived CoI presence and behaviorally enacted CoI presence differently. While prior CoI research has often relied on questionnaires to examine perceived CP, SP, and TP (Arbaugh et al., 2008; Stenbom, 2018), discourse-based approaches capture observable interaction and knowledge-construction processes (De Wever et al., 2006; Rourke et al., 1999). The present findings suggest that AI-supported environments may strengthen students' subjective sense of immediacy, support, and engagement without necessarily triggering more higher-order inquiry, peer interaction, or learner-enacted facilitation. Therefore, perceived and behaviorally enacted CoI presence should be treated as distinct indicators of online discussion quality in AI-mediated learning environments.

Practically, the findings suggest that AI TAs are valuable for routine and immediate support

but could not replace human facilitators in ADFs. AI TAs can provide timely responses, individualized feedback, emotional encouragement, and basic discourse support, but AI facilitation alone may be insufficient to foster deeper inquiry, reciprocal peer interaction, or learner-enacted facilitation in ADFs. This supports a human–AI collaborative facilitation model, in which AI TAs undertake routine facilitation tasks, while human TAs and instructors remain responsible for higher-level pedagogical orchestration, epistemic challenge, and relational support. This view is consistent with research showing that AI systems and educational chatbots can support routine instructional work but still require human oversight for reliability, accuracy, ethical use, and pedagogical alignment (Labadze et al., 2023; Zawacki-Richter et al., 2019) and with broader arguments that AI systems should be designed to support learner agency and human-centered pedagogical decision-making (Holstein & Aleven, 2022; Ouyang & Jiao, 2021).

Limitations and Future Research Directions

Several limitations should be acknowledged. First, this study adopted a quasi-experimental design with a fixed stage sequence, so the findings cannot support strong causal conclusions and may be affected by time, order, topic, course progression, or maturation effects. Future studies may use randomized or counterbalanced designs to better isolate AI TA effects. Second, participants were drawn from an Educational Technology program, which limits the generatability of the findings because students' AI literacy and prior experience may have shaped their perceptions and interaction behaviors (Chan & Hu, 2023; Laupichler et al., 2022; Long & Magerko, 2020). Further work could include more diverse disciplines and learner populations. Third, the study was conducted in a teacher-directed, topic-centered ADF, where

participation may have been shaped by task-completion norms rather than sustained inquiry. It would be interesting to examine AI TA support in other ADF tasks, such as student-led inquiry, project-based collaboration, and problem-solving discussions. Fourth, the study was implemented in a small private online course in a blended learning context. Future research may examine whether similar perception–behavior gaps emerge in MOOCs and other online discussion environments.

Conclusion

This 10-week explanatory mixed-methods study examined how students’ perceived and behaviorally enacted CoI presence in ADFs varied across TA conditions. The quantitative findings revealed a perception–behavior gap: students reported higher self-reported CP, SP, and TP during the AI-TA-supported stage, but these perceptions were not fully reflected in observable forum behaviors. Qualitative findings suggest that, compared with human TA support, the AI TA was perceived as advantageous in providing structured cognitive support, a low-pressure interaction atmosphere, and timely and accessible guidance, but limited in inquiry flexibility, emotional authenticity, and perceived authority. Overall, AI TAs may offer immediate and structured support, but AI-supported ADFs should adopt human–AI collaborative facilitation models to sustain deeper inquiry, peer interaction, and sense of accountability.

List of abbreviations

ADF: Asynchronous discussion forum

AI TA: Artificial intelligence teaching assistant

ATS: ANOVA-type statistic

685 CoI: Community of Inquiry

686 CP: Cognitive presence

687 RTE: Relative treatment effect

688 SP: Social presence

689 TA: Teaching assistant

690 TP: Teaching presence

691 **Declarations**

692 **Ethics approval and consent to participate**

693 Ethical approval was obtained from the Academic Committee of the School of Educational
694 Information Technology, South China Normal University (system application serial No.
695 1157202). Participants in the experimental cohort provided written informed consent before data
696 collection and were informed that participation was voluntary, that their data would be used
697 solely for research purposes, and that participation or non-participation would not affect their
698 course grades. The historical cohort data consisted solely of archived forum posts from the same
699 course and were retrospectively analyzed after de-identification with permission from the course
700 instructor.

701 **Consent for publication**

702 Not applicable. The manuscript does not contain identifiable personal information. All
703 student names and potentially identifiable information were removed or anonymized, and the
704 interaction example was anonymized using labels such as “Student 1.”

Availability of data and materials

The datasets generated and/or analyzed during the current study are not publicly available due to participant privacy and institutional restrictions. Anonymized data may be available from the corresponding author upon reasonable request and with permission from the relevant institution.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

YT contributed to conceptualization, AI TA design, data collection, data analysis, and manuscript writing. HL contributed to data collection and data analysis. LY and JL contributed to AI TA design and development, literature collection, and figure preparation. YZ contributed to conceptual guidance, supervision, manuscript revision, and overall study refinement. All authors read and approved the final manuscript.

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Declaration of generative AI

During the preparation of this manuscript, the authors used generative AI tools for language polishing, structural checking, and editing assistance. The authors reviewed and edited all AI-assisted outputs and take full responsibility for the content of the manuscript.

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